Construction of a Testable Prototype and Final Prototype:

Frame and Guillotine Door:

Our original prototype (which will be referred to as “the cardboard mockup”) for the frame and guillotine doors was constructed out of cardboard, popsicle sticks, hot glue, and masking tape, and was scaled down in size. The purpose of this cardboard mockup, in terms of the frame and guillotine door, was to help us properly visualize the frame's proportions and the guillotine door's functionality.

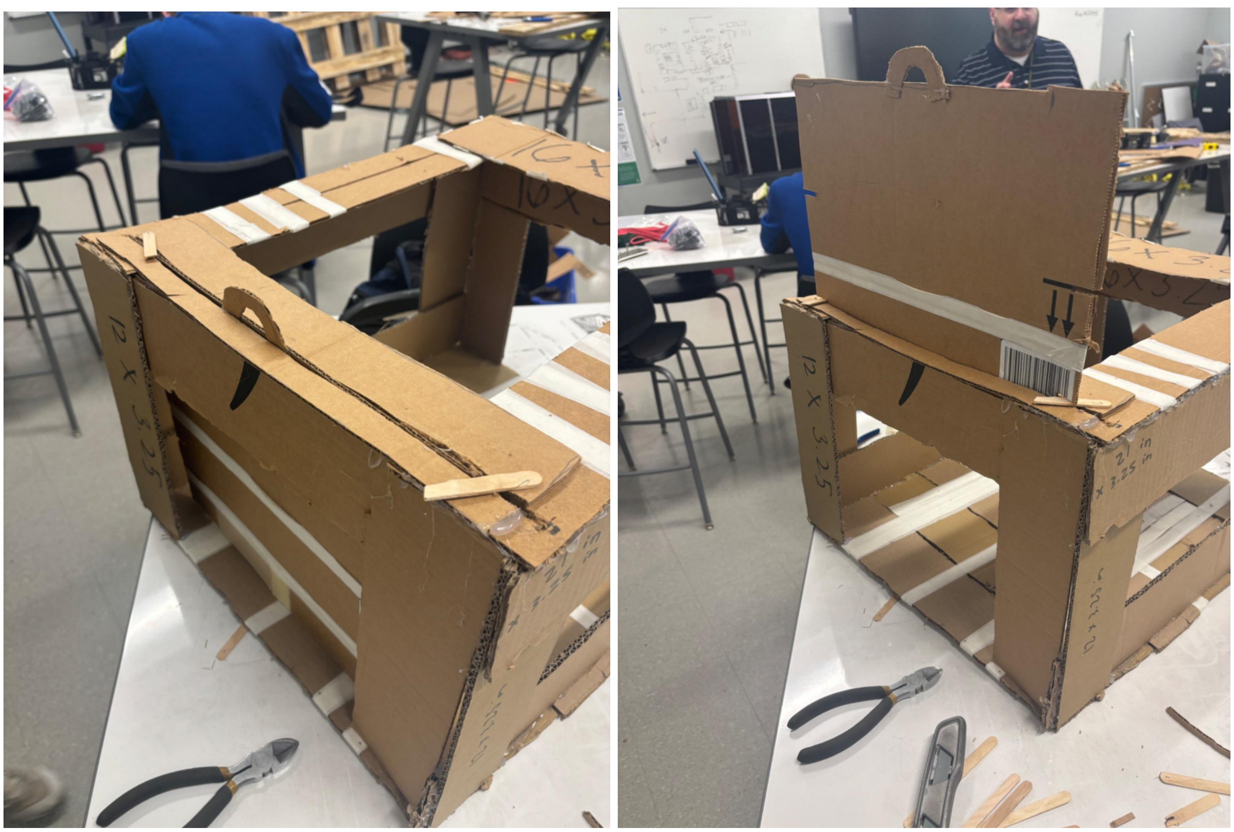
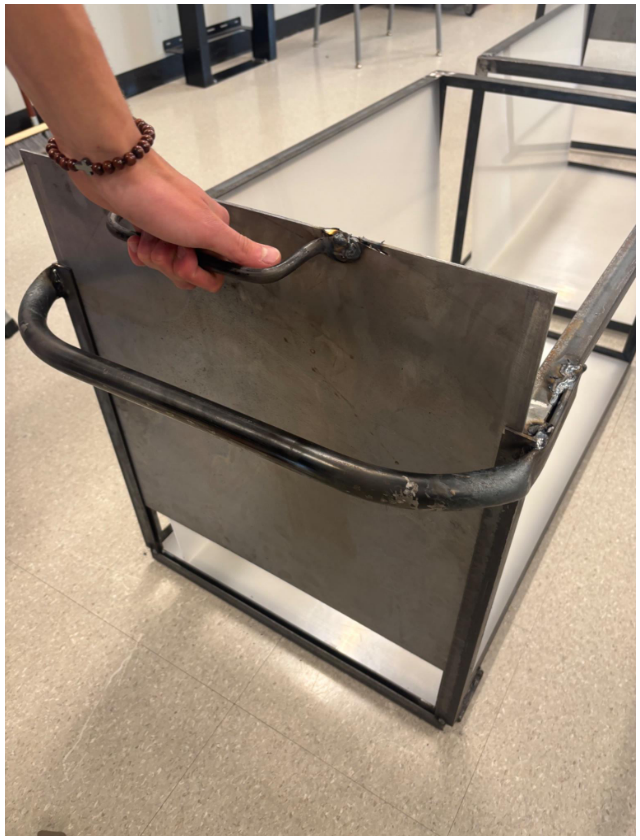


Photo of cardboard mockup

The final prototype’s frame is made of mild steel, providing durability and structural integrity. The frame consists of three boxes, each measuring 20’’ by 20’’ by 36’′. The steel frame is designed to be strong enough to support and contain the Komodo dragons securely. This frame was produced by another team to be connected by our team.

The guillotine door is a 19.75’’ by 19.75’’ sheet of mild steel that is 3/16 inches thick. The door slides vertically and is designed to allow a quick opening and closing motion. We determined that the locks used on the prototype were not necessary since it would require relatively considerable force to move this door upwards, meaning the dragon would have to push up against the door without good leverage to open it. We decided that the confidence in this qualitative data made mathematical modeling unnecessary since the data already demonstrated that the door would hold in place. The door was tested and can be removed within 1.5 seconds and inserted within 3 seconds by our team, which means that it satisfies the primary design requirements given previously.



The dimensions of the frame and door are designed to fit the containment system and facilitate smooth functionality. We limited the size of the frame due to the zookeeper needing to transport the Komodo Dragons through tight doorways from the Komodo Dragon Enclosure to the Medical Facility that measure only about 2 and a half feet wide. There are still minor adjustments to be made, such as reducing friction and correcting the tightness of the sliding gap, but the door is still functional despite this and satisfies the design requirements set out.

Box Connection System:

When creating the cardboard mockup for the frame and guillotine door, we also added small toggle clamps to visualize the functionality of the butterfly toggle clamps. These toggle were intended to hold the boxes together originally, however, after the mathematical modelling done in Element F, we determined that we would need to shift these downwards and use them primarily to hold back the torque of the boxes rather than to hold the boxes together in their entirety.

The box connection system has three distinct parts: Butterfly toggle clamps, PVC pipe connection system, and the extruding parts of the frame, which will be referred to as feet of the frame.

The frame's feet, which are thick rectangular prisms made of mild steel and welded to the bottom of the frame, do most of the heavy lifting. The feet are part of the frame and were created and welded at the same time as the frame.

The butterfly toggle clamps are designed to ensure that no gaps open between parts of the frame during the transportation of the box. They hold the boxes together primarily laterally so that the tendency of the first and last box to rotate is negated. The specific model of butterfly toggle clamps that we decided to use was the "Uxcell 1.61-inch Iron Butterfly Twist Latch Keeper Toggle Clamps for Case Box". The butterfly clamps are positioned along the side of the frame and attach to every vertical connection point of the frame. They are connected to the frame by a nut and bolt. Each butterfly clamp can withstand a force of around 45N-135N, which is suitable for our needs.

The last part of the overall connection system is the PVC pipe connection system, which was implemented to hold the boxes at the same elevation. The system consists of 1 ½ ” PVC pipes that are about 4” each, which are held down by grey PVC clamps. These clamps are placed at the connection areas on the top of each part of the frame. A Thinner PVC pipe is placed inside the larger PVC pipes to connect the system.

As demonstrated in Element F, this PVC is sturdy enough to hold the boxes together even with the Komodo dragon inside. We also tested the box to ensure that it would hold, and after placing 100-150 lbs inside the box, primarily distributed in the center (which has the highest chance of breaking the box), the box held without issue and could be moved or shaken without separating.

Materials and Construction Process:

The steel used for the prototype was sourced by Corbin Wolfgang, a member of the team, who obtained it from his father's shop. Corbin also handled the welding and cutting of the steel himself, ensuring precise measurements and high-quality craftsmanship. This hands-on approach allowed the team to maintain control over the accuracy and quality of the materials and construction. Each box in the prototype measures 20'' by 20'' by 36'', designed to accommodate the Komodo dragon comfortably while still meeting the constrained size. Once both the frame and doors were complete, both were painted with several layers of black Rust-Oleum paint primer.

At this stage, no additional materials will be added to enhance durability, and no safety mechanisms will be installed in the prototype.

Limitations and Areas for Improvement:

While the prototype has been constructed to meet most design requirements, there are still areas that need attention. The long-term durability of the box connection system, especially with repeated disassembly and reassembly, needs further evaluation. Additionally, the sliding guillotine door requires refinement to ensure it operates smoothly and consistently. Specifically, the door needs further adjustments to reduce friction and ensure it slides freely in the slot.